

Poisson SuSiE for One Count Vector

1 Model

We observe one count vector $y \in \mathbb{N}^M$ and a known nonnegative dictionary $F \in \mathbb{R}_+^{K \times M}$. The Poisson SuSiE model writes the Poisson rate as a sparse sum of D effects:

$$y_m \sim \text{Poi} \left(\sum_{d=1}^D \beta_d F_{\gamma_d m} \right), \quad m = 1, \dots, M.$$

For each effect d ,

$$\gamma_d \sim \text{Cat} \left(\frac{1}{K}, \dots, \frac{1}{K} \right), \quad (1)$$

$$\beta_d \mid \gamma_d \sim \text{Gam}(\alpha_d^0, \lambda_d^0), \quad (2)$$

where the Gamma distribution is parameterized by shape and rate. The index $\gamma_d \in \{1, \dots, K\}$ selects one row of F , and $\beta_d > 0$ is the effect size.

2 Variational Family

The approximation factorizes over effects:

$$q(\beta, \gamma) = \prod_{d=1}^D q(\gamma_d) q(\beta_d \mid \gamma_d),$$

with

$$q(\gamma_d = k) = \bar{\gamma}_{dk}, \quad q(\beta_d \mid \gamma_d = k) = \text{Gam}(\alpha_{dk}, \lambda_{dk}).$$

We also introduce auxiliary responsibilities $\xi_{dm} \geq 0$ satisfying $\sum_{d=1}^D \xi_{dm} = 1$ for each coordinate m . They allocate the count y_m across the D effects.

The moments used by the updates are

$$\mathbb{E}[\beta_d \mid \gamma_d = k] = \frac{\alpha_{dk}}{\lambda_{dk}}, \quad (3)$$

$$\mathbb{E}[\log \beta_d \mid \gamma_d = k] = \psi(\alpha_{dk}) - \log \lambda_{dk}, \quad (4)$$

$$\mathbb{E}[\log \beta_d] = \sum_{k=1}^K \bar{\gamma}_{dk} \{ \psi(\alpha_{dk}) - \log \lambda_{dk} \}, \quad (5)$$

$$\mathbb{E}[\log F_{\gamma_d m}] = \sum_{k=1}^K \bar{\gamma}_{dk} \log F_{km}. \quad (6)$$

3 ELBO

Using Jensen's inequality with the responsibilities ξ , the evidence lower bound is

$$\begin{aligned}\mathcal{L}(q) = & \sum_{d=1}^D \sum_{m=1}^M \xi_{dm} y_m \{ \mathbb{E}[\log \beta_d] + \mathbb{E}[\log F_{\gamma_d m}] - \log \xi_{dm} \} \\ & - \sum_{d=1}^D \sum_{k=1}^K \bar{\gamma}_{dk} \frac{\alpha_{dk}}{\lambda_{dk}} \sum_{m=1}^M F_{km} - \sum_{d=1}^D \text{KL}(q(\gamma_d, \beta_d) \parallel p(\gamma_d, \beta_d)).\end{aligned}\quad (7)$$

The KL term decomposes into discrete and continuous parts:

$$\text{KL}_{\text{disc}} = \sum_{d=1}^D \sum_{k=1}^K \bar{\gamma}_{dk} \left(\log \bar{\gamma}_{dk} - \log \frac{1}{K} \right), \quad (8)$$

$$\begin{aligned}\text{KL}_{\text{cont}} = & \sum_{d=1}^D \sum_{k=1}^K \bar{\gamma}_{dk} \left[(\alpha_{dk} - \alpha_d^0) \psi(\alpha_{dk}) + \alpha_d^0 (\log \lambda_{dk} - \log \lambda_d^0) \right. \\ & \left. - \log \frac{\Gamma(\alpha_{dk})}{\Gamma(\alpha_d^0)} - (\lambda_{dk} - \lambda_d^0) \frac{\alpha_{dk}}{\lambda_{dk}} \right].\end{aligned}\quad (9)$$

4 Coordinate Updates

Responsibility update. For each coordinate m ,

$$\xi_{dm} \propto \exp \left\{ \mathbb{E}[\log \beta_d] + \sum_{k=1}^K \bar{\gamma}_{dk} \log F_{km} \right\}, \quad d = 1, \dots, D,$$

normalized so that $\sum_d \xi_{dm} = 1$.

Gamma variational parameters. For effect d and factor k ,

$$\alpha_{dk} \leftarrow \alpha_d^0 + \sum_{m=1}^M \xi_{dm} y_m, \quad (10)$$

$$\lambda_{dk} \leftarrow \lambda_d^0 + \sum_{m=1}^M F_{km}. \quad (11)$$

The shape update does not depend on k because the same allocated count is used for every candidate factor.

Categorical assignment update. After dropping terms constant in k ,

$$\log \bar{\gamma}_{dk} \leftarrow \sum_{m=1}^M \xi_{dm} y_m \log F_{km} - \left(\alpha_d^0 + \sum_{m=1}^M \xi_{dm} y_m \right) \log \left(\lambda_d^0 + \sum_{m=1}^M F_{km} \right),$$

then $\bar{\gamma}_{d1}, \dots, \bar{\gamma}_{dK}$ are normalized by a softmax.

5 Empirical Bayes Hyperparameter Update

When `update_prior = TRUE`, the implementation updates $(\alpha_d^0, \lambda_d^0)$ for each effect. Define

$$\bar{\beta}_d = \sum_{k=1}^K \bar{\gamma}_{dk} \frac{\alpha_{dk}}{\lambda_{dk}}, \quad (12)$$

$$\overline{\log \beta}_d = \sum_{k=1}^K \bar{\gamma}_{dk} \{\psi(\alpha_{dk}) - \log \lambda_{dk}\}. \quad (13)$$

Maximizing the expected Gamma prior gives

$$\lambda_d^0 = \frac{\alpha_d^0}{\bar{\beta}_d}, \quad \psi(\alpha_d^0) = \overline{\log \beta}_d + \log \lambda_d^0.$$

The second equation is solved numerically by root finding in the current R implementation.

6 Algorithm

Algorithm 1 Poisson SuSiE

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1: Initialize  $\bar{\gamma}_{dk}$ ,  $\alpha_{dk}$ , and  $\lambda_{dk}$ .
2: for  $t = 1, \dots, T$  do
3:   if symmetry breaking is enabled then
4:     Reset duplicate active effects to the uniform assignment.
5:   end if
6:   for  $d = 1, \dots, D$  do
7:     Update  $\xi_{dm}$  for all  $d$  and  $m$  using the current posterior moments.
8:     Update  $\alpha_{dk}$  and  $\lambda_{dk}$  for the selected effect  $d$ .
9:     Update  $\bar{\gamma}_{dk}$  for the selected effect  $d$  by softmax normalization.
10:    if empirical Bayes is enabled then
11:      Update  $\alpha_d^0$  and  $\lambda_d^0$ .
12:    end if
13:  end for
14:  Evaluate the ELBO.
15: end for
```

7 Effect of Symmetry Breaking

When D is larger than the number of truly active components, multiple effects can converge to nearly identical posterior assignment vectors. The optional symmetry-breaking step detects such duplicate active effects, averages one copy with the duplicate, and resets the other copy to the uniform assignment over factors. In the simulation in the workflow analysis, this keeps the recovered active factors unchanged up to permutation, while making unused effects more diffuse instead of duplicating an already-explained factor. The tradeoff is that the reset is an explicit perturbation of coordinate ascent, so the ELBO can show small jumps around symmetry-breaking updates.